

Continuous-Time Signal Analysis: The Fourier Transform

Chapter 7

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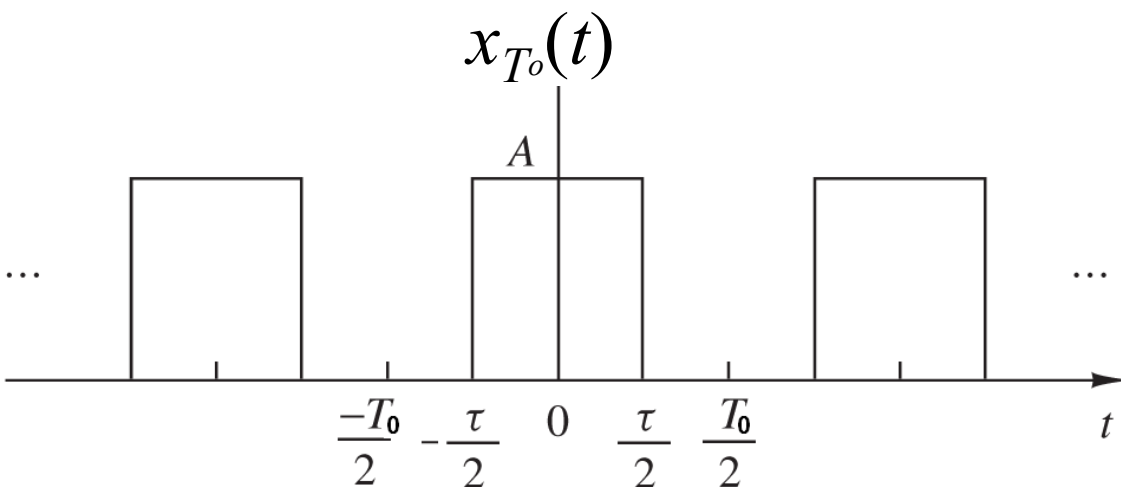
Chapter Outline

- Aperiodic Signal Representation by Fourier Integral
- Fourier Transform of Useful Functions
- Properties of Fourier Transform
- Signal Transmission Through LTIC Systems
- Ideal and Practical Filters
- Signal Energy
- Applications to Communications
- Data Truncation: Window Functions

Link between FT and FS

Fourier series (FS) allows us to represent periodic signal in term of sinusoidal or exponentials $e^{jn\omega_0 t}$.

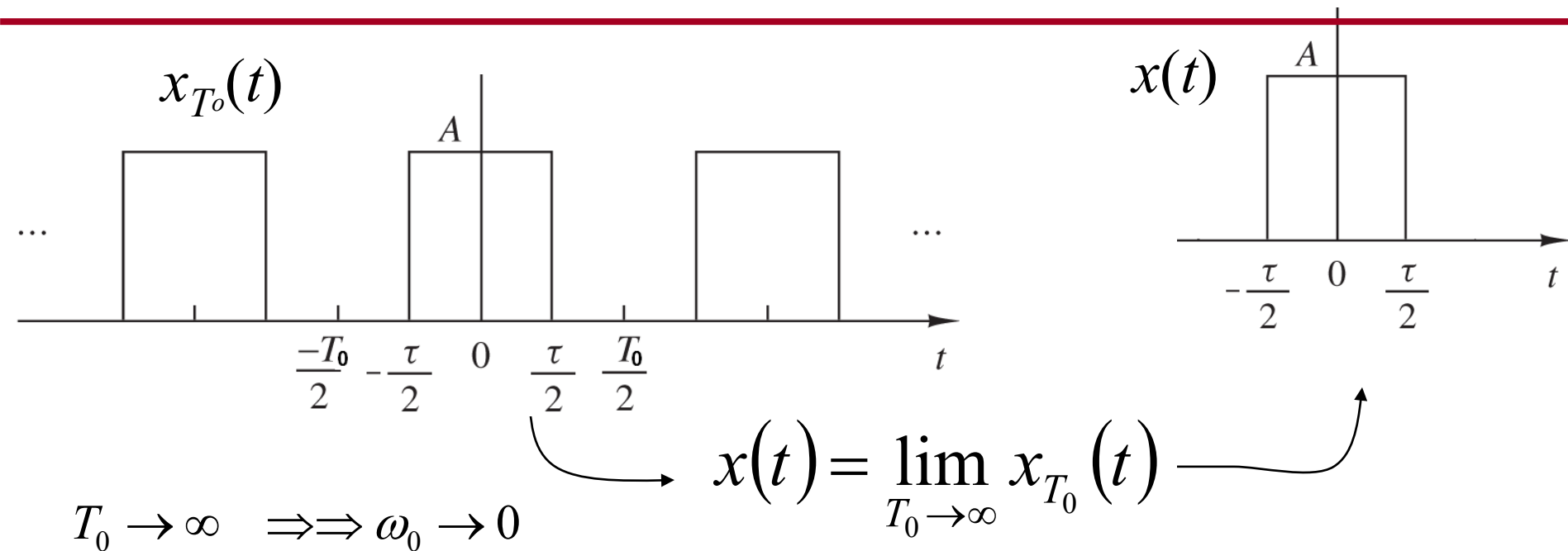
Fourier transform (FT) allows us to represent aperiodic (not periodic) signal in term of exponentials $e^{j\omega t}$.



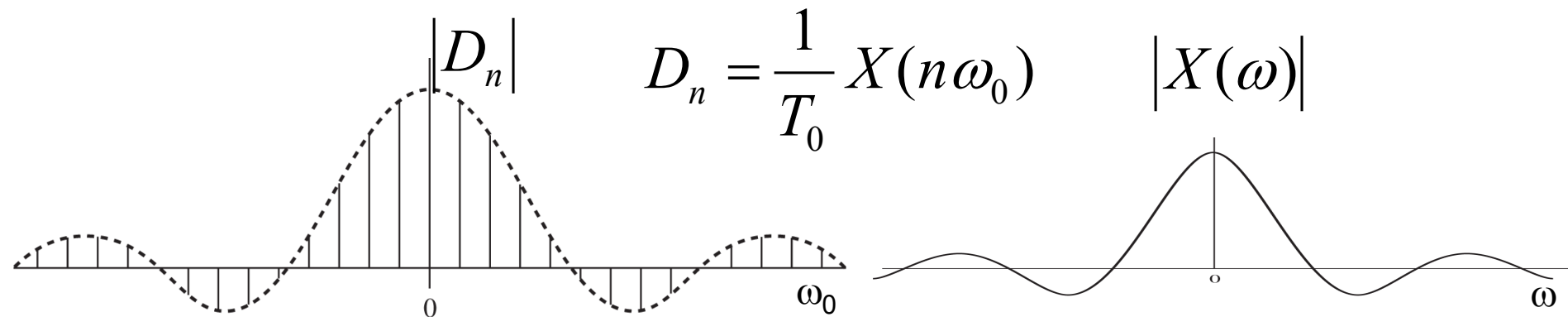
$$x_{T_0}(t) = \sum_{n=-\infty}^{\infty} D_n e^{jn\omega_0 t}$$

$$D_n = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} x_{T_0}(t) e^{-jn\omega_0 t} dt$$

Link between FT and FS



As T_0 gets larger and larger the fundamental frequency ω_0 gets smaller and smaller so the spectrum becomes continuous.



The Fourier Transform Spectrum

The Fourier transform:

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$X(\omega) = |X(\omega)| e^{j\angle X(\omega)}$$

The Amplitude (Magnitude) Spectrum

The Phase Spectrum

The amplitude spectrum is an even function and the phase is an odd function.

The Inverse Fourier transform:

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$$

Example

Find the Fourier transform of $x(t) = e^{-at}u(t)$, the magnitude, and the spectrum

Solution:

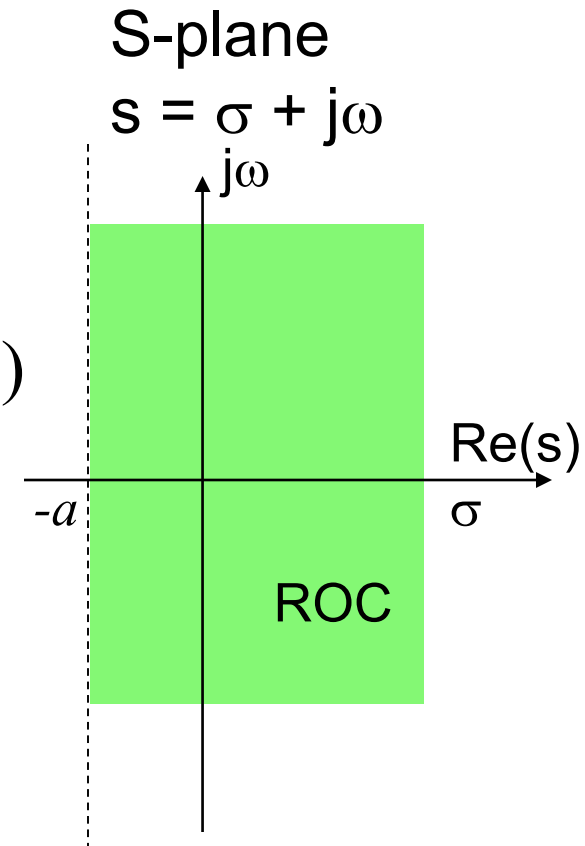
$$X(\omega) = \int_0^{\infty} e^{-at} e^{-j\omega t} dt = \frac{1}{a + j\omega} \quad \text{if } a > 0$$

$$|X(\omega)| = \frac{1}{\sqrt{a^2 + \omega^2}} \quad \angle X(\omega) = -\tan^{-1}(\omega / a)$$

How does $X(\omega)$ relates to $X(s)$?

$$X(s) = \int_0^{\infty} e^{-at} e^{-st} dt = -\frac{1}{a + s} e^{-(s+a)t} \bigg|_0^{\infty}$$

$$X(s) = \frac{1}{a + s} \quad \text{if } \operatorname{Re}(s) > -a$$

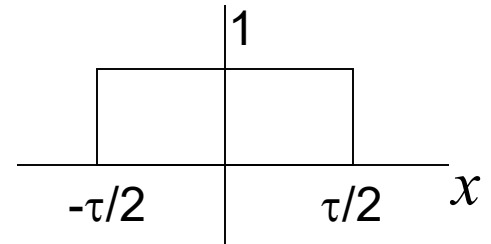


Since the $j\omega$ -axis is in the region of convergence then FT exist.

Useful Functions

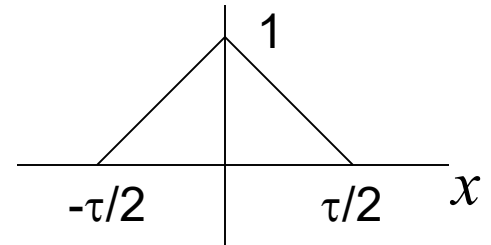
Unit Gate Function

$$\text{rect}\left(\frac{x}{\tau}\right) = \begin{cases} 0 & |x| > \tau/2 \\ 0.5 & |x| = \tau/2 \\ 1 & |x| < \tau/2 \end{cases}$$



Unit Triangle Function

$$\Delta\left(\frac{x}{\tau}\right) = \begin{cases} 0 & |x| \geq \tau/2 \\ 1 - (2/\tau)|x| & |x| < \tau/2 \end{cases}$$



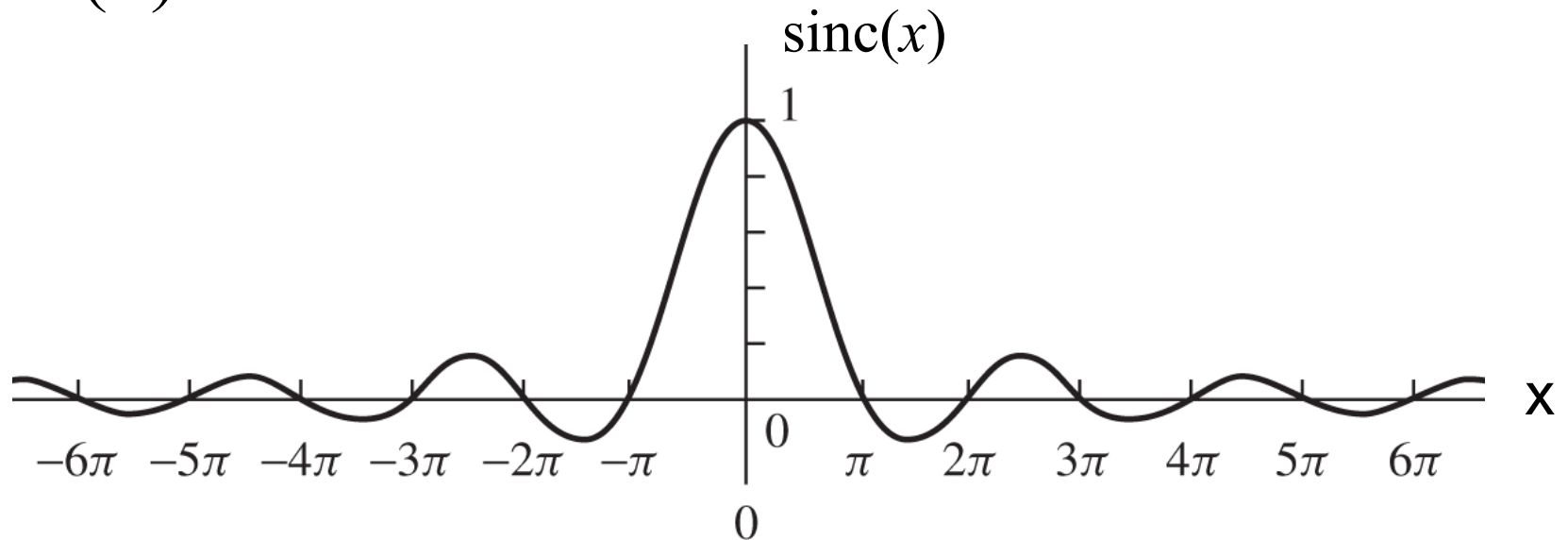
Useful Functions

Interpolation Function

$$\text{sinc}(x) = \frac{\sin x}{x}$$

$$\text{sinc}(x) = 0 \quad \text{for } x = \pm k\pi$$

$$\text{sinc}(x) = 1 \quad \text{for } x = 0$$



Example

Find the FT, the magnitude, and the phase spectrum of $x(t) = \text{rect}(t/\tau)$.

Answer

$$X(\omega) = \int_{-\tau/2}^{\tau/2} \text{rect}(t/\tau) e^{-j\omega t} dt = \tau \text{sinc}(\omega\tau/2)$$

What is the bandwidth of the above pulse?

The spectrum of a pulse extend from 0 to ∞ . However, much of the spectrum is concentrated within the first lobe ($\omega=0$ to $2\pi/\tau$)

Examples

Find the FT of the unit impulse $\delta(t)$.

Answer

$$X(\omega) = \int_{-\infty}^{\infty} \delta(t) e^{-j\omega t} dt = 1 \qquad \delta(t) \leftrightarrow 1$$

Find the inverse FT of $\delta(\omega)$.

Answer

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \delta(\omega) e^{j\omega t} d\omega = \frac{1}{2\pi} \qquad 1 \leftrightarrow 2\pi\delta(\omega)$$

Examples

Find the inverse FT of $\delta(\omega - \omega_0)$.

Answer

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \delta(\omega - \omega_0) e^{j\omega t} d\omega = \frac{1}{2\pi} e^{j\omega_0 t}$$

so the spectrum of a complex exponent is a shifted impulse

$$e^{j\omega_0 t} \leftrightarrow 2\pi\delta(\omega - \omega_0) \quad \text{and} \quad e^{-j\omega_0 t} \leftrightarrow 2\pi\delta(\omega + \omega_0)$$

Find the FT of the everlasting sinusoid $\cos(\omega_0 t)$.

Answer

$$\cos \omega_0 t = \frac{1}{2} (e^{j\omega_0 t} + e^{-j\omega_0 t})$$

$$\frac{1}{2} (e^{j\omega_0 t} + e^{-j\omega_0 t}) \leftrightarrow \pi [\delta(\omega + \omega_0) + \delta(\omega - \omega_0)]$$

Examples

Find the FT of a periodic signal.

Answer

$$x(t) = \sum_{n=-\infty}^{n=\infty} D_n e^{jn\omega_0 t} \quad \omega_0 = 2\pi / T_0$$

Take the FT of both side and use linearity property of FT

$$X(\omega) = 2\pi \sum_{n=-\infty}^{n=\infty} D_n \delta(\omega - n\omega_0)$$

Examples

Find the FT of the unit impulse train $\delta_{T_0}(t)$

Answer

$$\delta_{T_0}(t) = \frac{1}{T_0} \sum_{n=-\infty}^{n=\infty} e^{jn\omega_0 t}$$

$$e^{j\omega_0 t} \leftrightarrow 2\pi\delta(\omega - \omega_0)$$

$$X(\omega) = \frac{2\pi}{T_0} \sum_{n=-\infty}^{n=\infty} \delta(\omega - n\omega_0)$$

TABLE 7.1 Fourier Transforms

No.	$x(t)$	$X(\omega)$	
1	$e^{-at}u(t)$	$\frac{1}{a + j\omega}$	$a > 0$
2	$e^{at}u(-t)$	$\frac{1}{a - j\omega}$	$a > 0$
3	$e^{-a t }$	$\frac{2a}{a^2 + \omega^2}$	$a > 0$
4	$te^{-at}u(t)$	$\frac{1}{(a + j\omega)^2}$	$a > 0$
5	$t^n e^{-at}u(t)$	$\frac{n!}{(a + j\omega)^{n+1}}$	$a > 0$
6	$\delta(t)$	1	
7	1	$2\pi\delta(\omega)$	
8	$e^{j\omega_0 t}$	$2\pi\delta(\omega - \omega_0)$	
9	$\cos \omega_0 t$	$\pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$	
10	$\sin \omega_0 t$	$j\pi[\delta(\omega + \omega_0) - \delta(\omega - \omega_0)]$	
11	$u(t)$	$\pi\delta(\omega) + \frac{1}{j\omega}$	
12	$\operatorname{sgn} t$	$\frac{2}{j\omega}$	

TABLE 7.1 Fourier Transforms

No.	$x(t)$	$X(\omega)$	
13	$\cos \omega_0 t u(t)$	$\frac{\pi}{2} [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)] + \frac{j\omega}{\omega_0^2 - \omega^2}$	
14	$\sin \omega_0 t u(t)$	$\frac{\pi}{2j} [\delta(\omega - \omega_0) - \delta(\omega + \omega_0)] + \frac{\omega_0}{\omega_0^2 - \omega^2}$	
15	$e^{-at} \sin \omega_0 t u(t)$	$\frac{\omega_0}{(a + j\omega)^2 + \omega_0^2}$	$a > 0$
16	$e^{-at} \cos \omega_0 t u(t)$	$\frac{a + j\omega}{(a + j\omega)^2 + \omega_0^2}$	$a > 0$
17	$\text{rect}\left(\frac{t}{\tau}\right)$	$\tau \text{sinc}\left(\frac{\omega\tau}{2}\right)$	
18	$\frac{W}{\pi} \text{sinc}(Wt)$	$\text{rect}\left(\frac{\omega}{2W}\right)$	
19	$\Delta\left(\frac{t}{\tau}\right)$	$\frac{\tau}{2} \text{sinc}^2\left(\frac{\omega\tau}{4}\right)$	
20	$\frac{W}{2\pi} \text{sinc}^2\left(\frac{Wt}{2}\right)$	$\Delta\left(\frac{\omega}{2W}\right)$	
21	$\sum_{n=-\infty}^{\infty} \delta(t - nT)$	$\omega_0 \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_0)$	$\omega_0 = \frac{2\pi}{T}$
22	$e^{-t^2/2\sigma^2}$	$\sigma\sqrt{2\pi} e^{-\sigma^2\omega^2/2}$	

Properties of the Fourier Transform

- *Linearity:*

Let $x(t) \Leftrightarrow X(\omega)$ *and* $y(t) \Leftrightarrow Y(\omega)$

then $\alpha x(t) + \beta y(t) \Leftrightarrow \alpha X(\omega) + \beta Y(\omega)$

- *Time Scaling:*

Let $x(t) \Leftrightarrow X(\omega)$

then $x(at) \Leftrightarrow \frac{1}{|a|} X\left(\frac{\omega}{a}\right)$

When $a > 1$ that leads to compression in the time domain which results in expansion in the frequency domain

Internet channel A can transmit 100k pulse/sec and channel B can transmit 200k pulse/sec. **Which channel does require higher bandwidth?**

Properties of the Fourier Transform

- Time Reversal:*

Let $x(t) \Leftrightarrow X(\omega)$

then $x(-t) \Leftrightarrow X(-\omega)$

Example: Find the FT of $e^{-a|t|}$

- Left or Right Shift in Time:*

Let $x(t) \Leftrightarrow X(\omega)$

Time shift effects the phase and not the magnitude.

then $x(t - t_0) \Leftrightarrow X(\omega)e^{-j\omega t_0}$

Example: Find the FT of $e^{-a|t-t_0|}$ and draw its magnitude and spectrum

Properties of the Fourier Transform

- Multiplication by a Complex Exponential (Freq. Shift Property):*

Let $x(t) \Leftrightarrow X(\omega)$

then $x(t)e^{j\omega_0 t} \Leftrightarrow X(\omega - \omega_0)$

- Multiplication by a Sinusoid (Amplitude Modulation):*

Let $x(t) \Leftrightarrow X(\omega)$

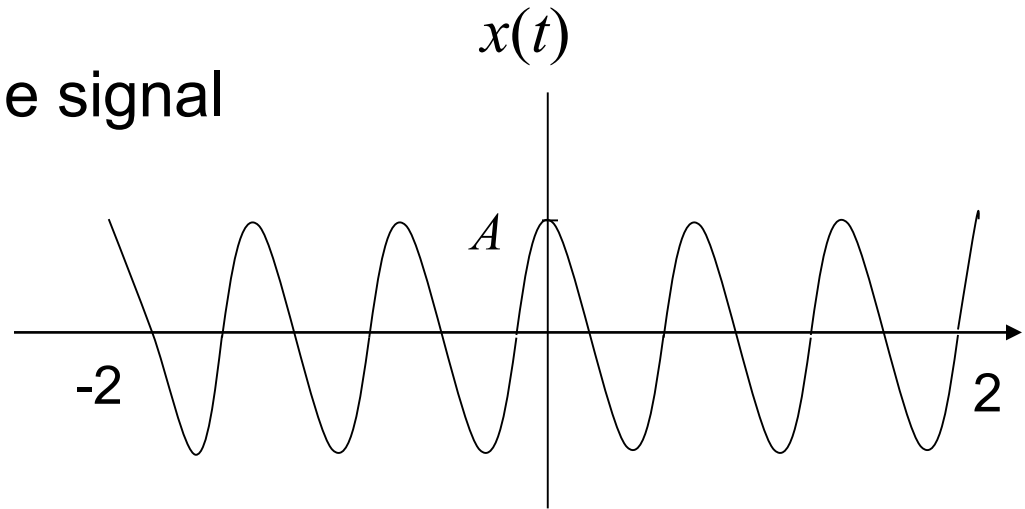
then $x(t)\cos(\omega_0 t) \Leftrightarrow \frac{1}{2}[X(\omega + \omega_0) + X(\omega - \omega_0)]$

$\cos \omega_0 t$ is the **carrier**, $x(t)$ is the modulating signal (**message**),
 $x(t) \cos \omega_0 t$ is the **modulated signal**.

Example: Amplitude Modulation

Example: Find the FT for the signal

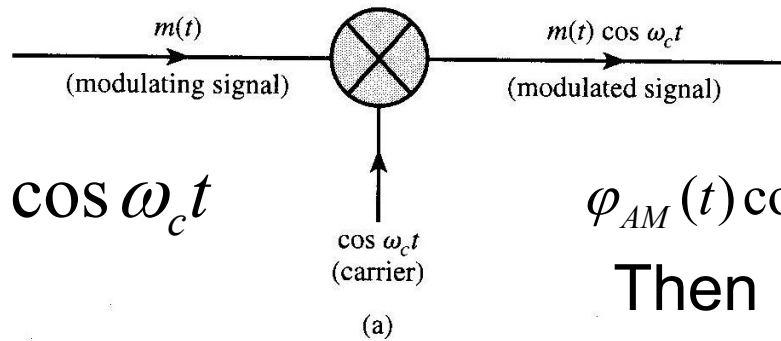
$$x(t) = \text{rect}(t / 4) \cos 10t$$



Amplitude Modulation

Modulation

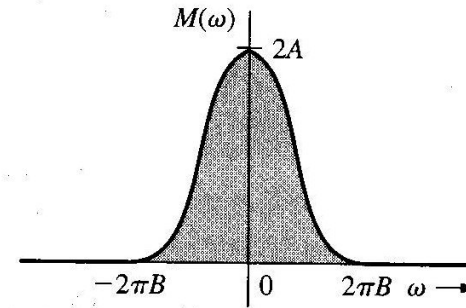
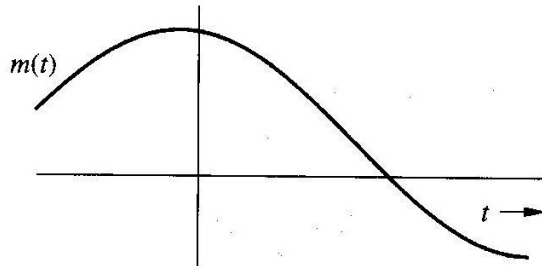
$$\varphi_{AM}(t) = m(t) \cos \omega_c t$$



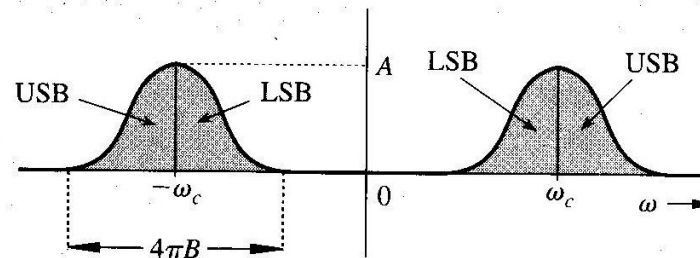
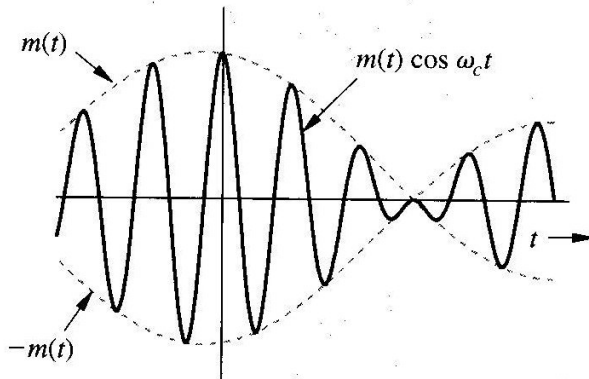
Demodulation

$$\varphi_{AM}(t) \cos \omega_c t = 0.5m(t)[1 + \cos 2\omega_c t]$$

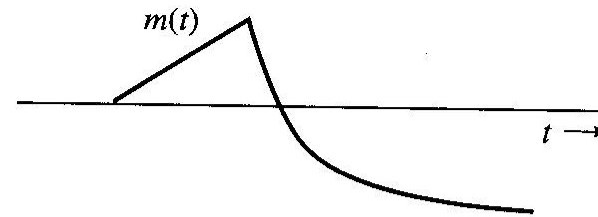
Then lowpass filtering



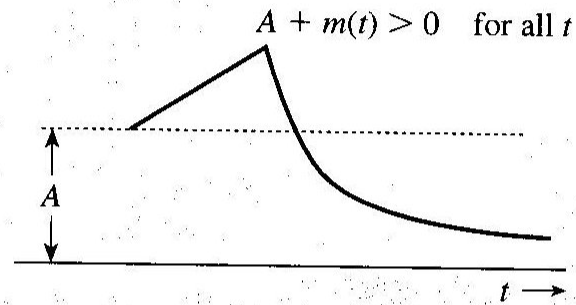
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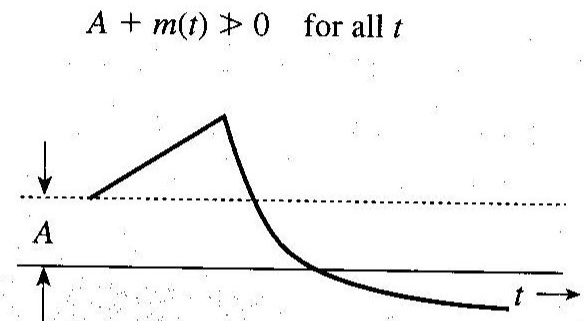
Amplitude Modulation: Envelope Detector



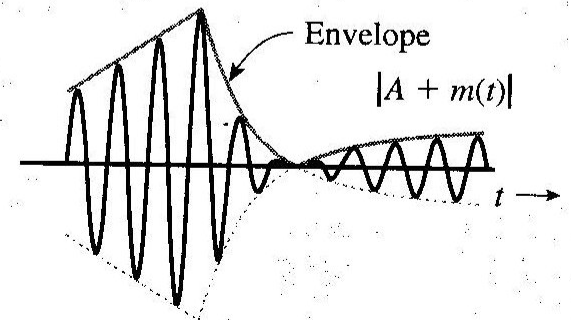
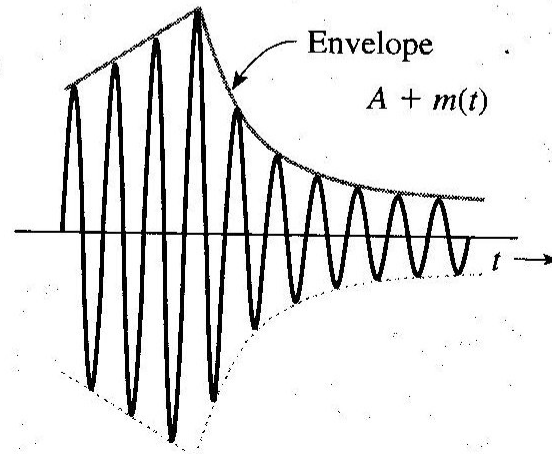
(a)



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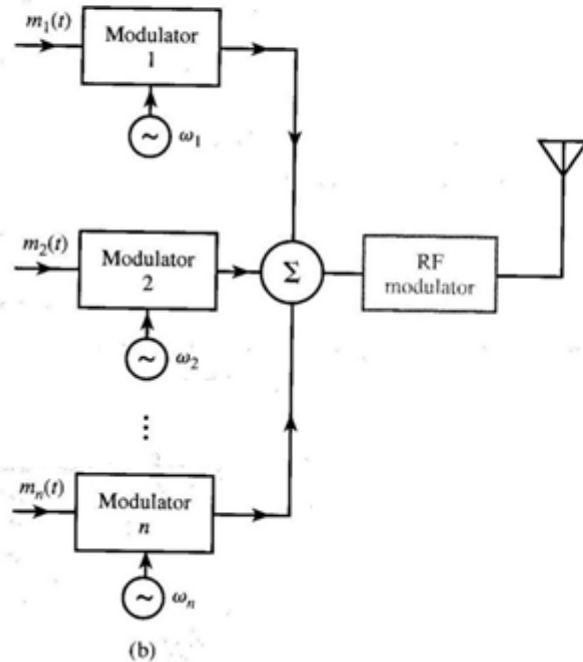
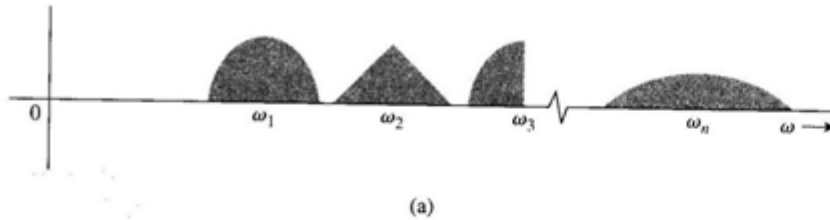


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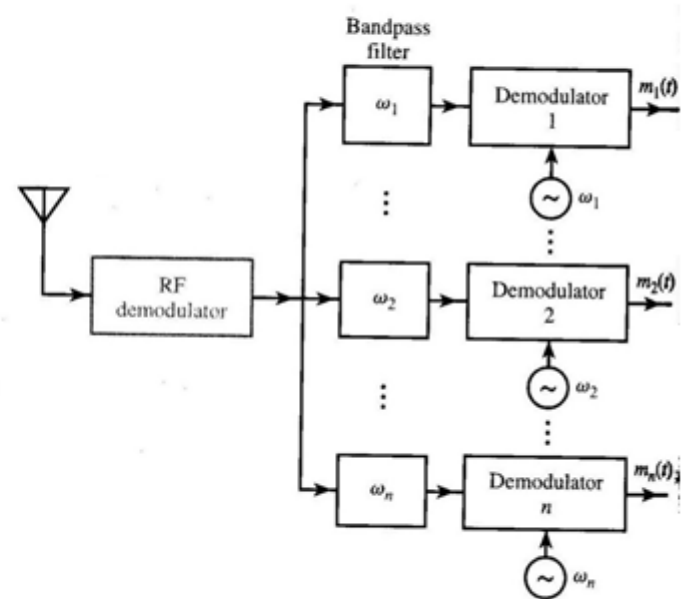


Applic. of Modulation: Frequency-Division Multiplexing

- 1- Transmission of different signals over different bands
- 2- Require smaller antenna



Transmitter



Receiver

Properties of the Fourier Transform

- *Differentiation in the Frequency Domain:*

- *Let* $x(t) \Leftrightarrow X(\omega)$

then $t^n x(t) \leftrightarrow (j)^n \frac{d^n}{d\omega^n} X(\omega)$

- *Differentiation in the Time Domain:*

Let $x(t) \Leftrightarrow X(\omega)$

then $\frac{d^n}{dt^n} x(t) \leftrightarrow (j\omega)^n X(\omega)$

Example: Use the time-differentiation property to find the Fourier Transform of the triangle pulse $x(t) = \Delta(t/\tau)$

Properties of the Fourier Transform

- *Integration in the Time Domain:*

Let $x(t) \Leftrightarrow X(\omega)$

Then $\int_{-\infty}^t x(\tau) d\tau \Leftrightarrow \frac{1}{j\omega} X(\omega) + \pi X(0) \delta(\omega)$

- *Convolution and Multiplication in the Time Domain:*

Let $x(t) \Leftrightarrow X(\omega)$
 $y(t) \Leftrightarrow Y(\omega)$

Then $x(t) * y(t) \Leftrightarrow X(\omega) Y(\omega)$

$$x_1(t) x_2(t) \Leftrightarrow \frac{1}{2\pi} X_1(\omega) * X_2(\omega) \quad \text{Frequency convolution}$$

Example

Find the system response to the input $x(t) = e^{-at} u(t)$ if the system impulse response is $h(t) = e^{-bt} u(t)$.

Properties of the Fourier Transform

- *Duality (Similarity) :*

- *Let* $x(t) \Leftrightarrow X(\omega)$

then $X(t) \Leftrightarrow 2\pi x(-\omega)$

$x(t)$	$X(\omega)$
$\text{rect}\left(\frac{t}{\tau}\right)$	$\tau \text{sinc}\left(\frac{\omega\tau}{2}\right)$

Energy of a Signal

- *Parseval's Theorem*: since $x(t)$ is non-periodic and has FT $X(\omega)$, then it is an energy signals:

$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(\omega)|^2 d\omega$$

Real signal has even spectrum $X(\omega) = X(-\omega)$, $E = \frac{1}{\pi} \int_0^{\infty} |X(\omega)|^2 d\omega$

Example

Find the energy of signal $x(t) = e^{-at} u(t)$. Determine the frequency ω so that the energy contributed by the spectrum components of all frequencies below ω is 95% of the signal energy E_X .

Answer: $\omega = 12.7a$ rad/sec

TABLE 7.2 Fourier Transform Operations

Operation	$x(t)$	$X(\omega)$
Scalar multiplication	$kx(t)$	$kX(\omega)$
Addition	$x_1(t) + x_2(t)$	$X_1(\omega) + X_2(\omega)$
Conjugation	$x^*(t)$	$X^*(-\omega)$
Duality	$X(t)$	$2\pi x(-\omega)$
Scaling (a real)	$x(at)$	$\frac{1}{ a } X\left(\frac{\omega}{a}\right)$
Time shifting	$x(t - t_0)$	$X(\omega)e^{-j\omega t_0}$
Frequency shifting (ω_0 real)	$x(t)e^{j\omega_0 t}$	$X(\omega - \omega_0)$
Time convolution	$x_1(t) * x_2(t)$	$X_1(\omega)X_2(\omega)$
Frequency convolution	$x_1(t)x_2(t)$	$\frac{1}{2\pi} X_1(\omega) * X_2(\omega)$
Time differentiation	$\frac{d^n x}{dt^n}$	$(j\omega)^n X(\omega)$
Time integration	$\int_{-\infty}^t x(u) du$	$\frac{X(\omega)}{j\omega} + \pi X(0)\delta(\omega)$

Data Truncation: Window Functions

- 1- Truncate $x(t)$ to reduce numerical computation
- 2- Truncate $h(t)$ to make the system response finite and causal
- 3- Truncate $X(\omega)$ to prevent aliasing in sampling the signal $x(t)$
- 4- Truncate D_n to synthesis the signal $x(t)$ from few harmonics.

$$x_w(t) = x(t)w(t) \quad \text{and} \quad X_w(\omega) = \frac{1}{2\pi} X(\omega) * W(\omega)$$

The diagram illustrates the time-domain equation $x_w(t) = x(t)w(t)$. Three upward-pointing arrows are positioned below the terms: the leftmost arrow points to $x_w(t)$ and is labeled 'Truncated Signal'; the middle arrow points to $x(t)$ and is labeled 'Original Signal'; the rightmost arrow points to $w(t)$ and is labeled 'Truncation Window'.

What are the implications of data truncation?

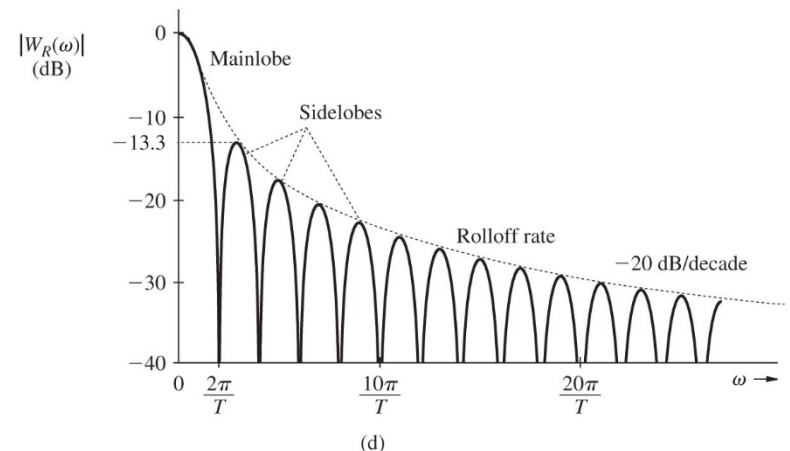
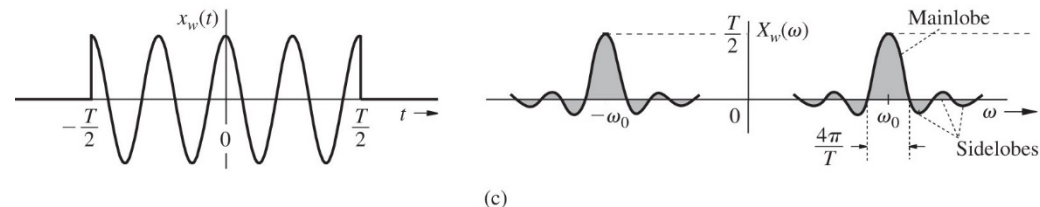
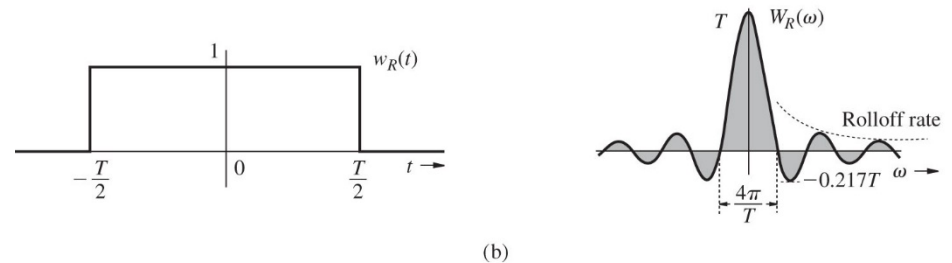
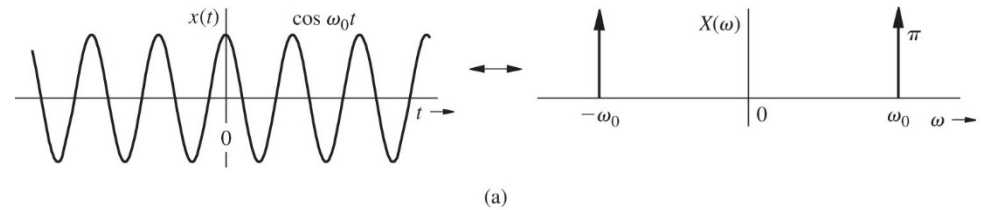
Implications of Data Truncation

- 1- Spectral spreading
- 2- Spectral leakage
- 3- Poor frequency resolution

What happened if $x(t)$ has two spectral components of frequencies differing by less than $4\pi/T$ rad/s ($2/T$ Hz)?

The ideal window for truncation is the one that has

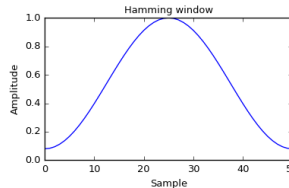
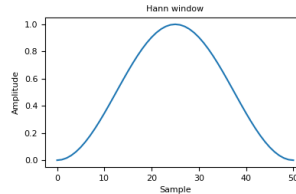
- 1- Smaller mainlobe width
- 2- Sidelobe with high rolloff rate
- 3- Small sidelobe peak



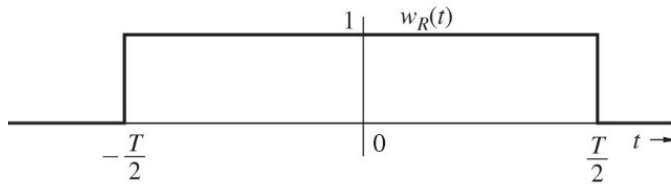
Data Truncation: Window Functions

Some Window Functions and Their Characteristics

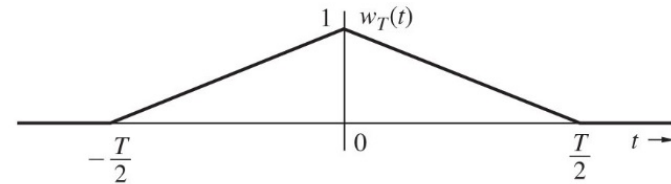
Window $w(t)$	Mainlobe Width	Rolloff Rate (dB/oct)	Peak Sidelobe level (dB)
Rectangular: $\text{rect}\left(\frac{t}{T}\right)$	$\frac{4\pi}{T}$	-6	-13.3
Bartlett: $\Delta\left(\frac{t}{2T}\right)$	$\frac{8\pi}{T}$	-12	-26.5
Hanning: $0.5 \left[1 + \cos\left(\frac{2\pi t}{T}\right) \right]$	$\frac{8\pi}{T}$	-18	-31.5
Hamming: $0.54 + 0.46 \cos\left(\frac{2\pi t}{T}\right)$	$\frac{8\pi}{T}$	-6	-42.7
Blackman: $0.42 + 0.5 \cos\left(\frac{2\pi t}{T}\right) + 0.08 \cos\left(\frac{4\pi t}{T}\right)$	$\frac{12\pi}{T}$	-18	-58.1
Kaiser: $\frac{I_0 \left[\alpha \sqrt{1 - 4 \left(\frac{t}{T} \right)^2} \right]}{I_0(\alpha)} \quad 0 \leq \alpha \leq 10$	$\frac{11.2\pi}{T}$	-6	-59.9 ($\alpha = 8.168$)



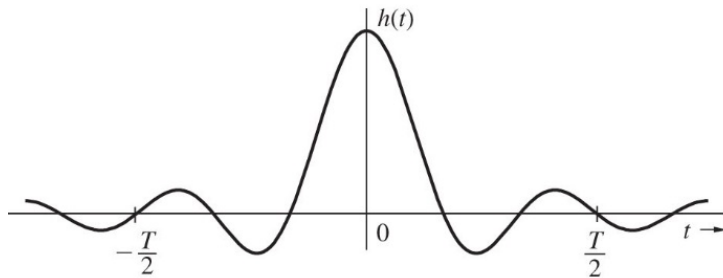
Using Windows in Filter Design



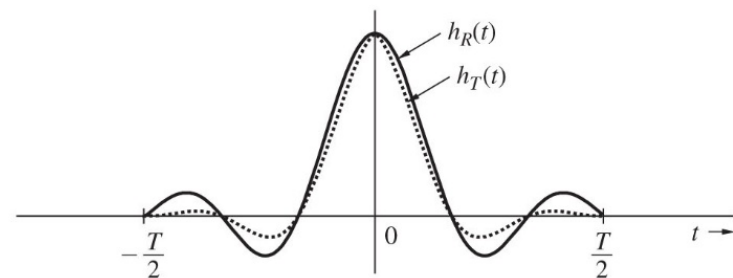
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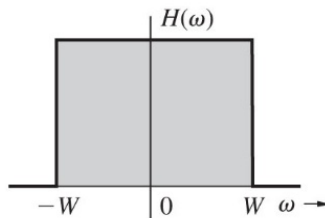
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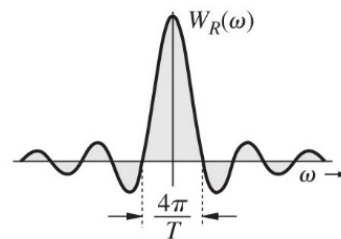
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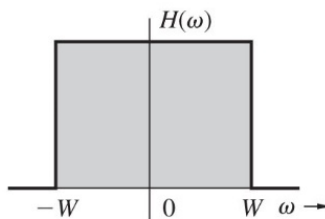
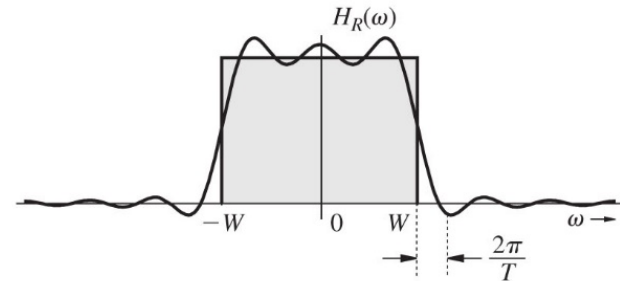


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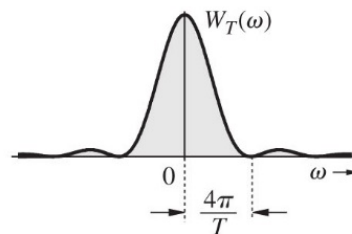


(e)

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(f)

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